COSC 32133-LAB2

PS/2020/010

Controller

StudentController.java

```
package com.example.lab2.Controller;
import com.example.lab2.Model.Student;
import com.example.lab2.Service.StudentService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@RestController
@RequestMapping("/api/students")
@CrossOrigin(origins = "*")
public class StudentController {

@Autowired
private StudentService studentService;
@PostMapping
public ResponseEntity<Student> saveEmployee(@RequestBody Student student){
    return new
ResponseEntity<Student>(studentService.saveStudent(student),HttpStatus.CREATED);
    }
     //GetAll Rest Api
    @GetMapping
    public List<Student> getAllStudents(){
    return studentService.getAllStudents();
    }
     //Get by Id Rest Api
    @GetMapping("{id}")
     // localhost:8080/api/employees/1
    public ResponseEntity<Student> getEmployeeById(@PathVariable("id") long
    employeeID){
    return new
ResponseEntity<Student>(studentService.getStudentById(employeeID),HttpStatus.OK);
    }
     //Update Rest Api
    @PutMapping("{id}")
    public ResponseEntity<Student> updateEmployee(@PathVariable("id") long
    id,
    @RequestBody Student employee){
    return new
ResponseEntity<Student>(studentService.updateStudent(employee,id),HttpStatus.OK);
```

```
    }
     //Delete Rest Api
    @DeleteMapping("{id}")
    public ResponseEntity<String> deleteStudent(@PathVariable("id") long
    id){
     //delete employee from db
    studentService.deleteStudent(id);
    return new ResponseEntity<String>("Student deleted
Successfully.",HttpStatus.OK);
    }

}
```

Model

Student.java

```
package com.example.lab2.Model;

import jakarta.persistence.*;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Entity
@AllArgsConstructor
@NoArgsConstructor
@Data
@Table(name="Student")
public class Student {
@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)
private long id;
private String firstName;
private String lastName;
private String email;
private String department;
private String yearOfEnrollment;
```

```java
public long getId() {
    return id;
}

public void setId(long id) {
    this.id = id;
}

// Getter and Setter for firstName
public String getFirstName() {
    return firstName;
}

public void setFirstName(String firstName) {
    this.firstName = firstName;
}

public String getLastName() {
    return lastName;
}

public void setLastName(String lastName) {
    this.lastName = lastName;
}

// Getter and Setter for email
public String getEmail() {
    return email;
}

public void setEmail(String email) {
    this.email = email;
}

// Getter and Setter for department
public String getDepartment() {
    return department;
}

public void setDepartment(String department) {
    this.department = department;
}

// Getter and Setter for yearOfEnrollment
```

```
public String getYearOfEnrollment() {
    return yearOfEnrollment;
}

public void setYearOfEnrollment(String yearOfEnrollment) {
    this.yearOfEnrollment = yearOfEnrollment;
}

}
```

Repository

StudentRepository.java

```
package com.example.lab2.Repository;
import org.springframework.data.jpa.repository.JpaRepository;

import com.example.lab2.Model.Student;

public interface StudentRepository extends JpaRepository<Student,Long> {

}
```

Service

Service.java

```
package com.example.lab2.Service.Impl;
import com.example.lab2.Model.Student;
import com.example.lab2.Repository.StudentRepository;
import com.example.lab2.Service.StudentService;

import java.util.List;
import java.util.Optional;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

@Service
public class StudentServiceImpl implements StudentService {
    @Autowired
```

```
    private StudentRepository studentRepository;

    @Override
    public Student saveStudent(Student student) {
        return studentRepository.save(student);
    }

    @Override
    public List<Student> getAllStudents() {
        // TODO Auto-generated method stub
        return studentRepository.findAll();
    }

    @Override
    public Student getStudentById(long id) {
        Optional<Student> employee = studentRepository.findById(id);
        if(employee.isPresent()){
        return employee.get();
        }else {
        throw new RuntimeException();
        }
    }


@Override
    public Student updateStudent(Student student, long id) {
    Student existingStudent =studentRepository.findById(id).orElseThrow(()-> new
RuntimeException());
existingStudent.setFirstName(student.getFirstName());
existingStudent.setLastName(student.getLastName());
existingStudent.setEmail(student.getEmail());
existingStudent.setDepartment(student.getDepartment());
existingStudent.setYearOfEnrollment(student.getYearOfEnrollment());
 // save
studentRepository.save(existingStudent);
return existingStudent;

    }

    @Override
    public void deleteStudent(long id) {
        studentRepository.findById(id).orElseThrow(()-> new
        RuntimeException());
         //delete
        studentRepository.deleteById(id);}
```

```
    }
```

Impl

StudentServiceImpl.java

```
package com.example.lab2.Service;
import java.util.List;

import com.example.lab2.Model.Student;

public interface StudentService {
Student saveStudent(Student student);
List<Student> getAllStudents();
Student getStudentById(long id);
Student updateStudent(Student student,long id);
void deleteStudent(long id);
}
```

Git Operations

```
D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git init
Initialized empty Git repository in D:/Users/jayasek-ps20010/Desktop/PS.2020.010-Lab sheet 2/lab2/.git/

D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git add .
warning: in the working copy of '.gitattributes', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.mvn/wrapper/maven-wrapper.properties', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'mvnw.cmd', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/lab2/Controller/StudentController.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/lab2/Lab2Application.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/lab2/Model/Student.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/lab2/Repository/StudentRepository.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/lab2/Service/Impl/StudentServiceImpl.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/lab2/Service/StudentService.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/resources/application.properties', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/test/java/com/example/lab2/Lab2ApplicationTests.java', LF will be replaced by CRLF the next time Git touches it
```

```
D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git commit -m "first commit"
[master (root-commit) 222456f] first commit
 Committer: PS/2020/010 - JAYASEKARA A.E.P.P <jayasek-ps20010@kln.ac.lk>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author
```

```
14 files changed, 790 insertions(+)
create mode 100644 .gitattributes
create mode 100644 .gitignore
create mode 100644 .mvn/wrapper/maven-wrapper.properties
create mode 100644 mvnw
create mode 100644 mvnw.cmd
create mode 100644 pom.xml
create mode 100644 src/main/java/com/example/lab2/Controller/StudentController.java
create mode 100644 src/main/java/com/example/lab2/Lab2Application.java
create mode 100644 src/main/java/com/example/lab2/Model/Student.java
create mode 100644 src/main/java/com/example/lab2/Repository/StudentRepository.java
create mode 100644 src/main/java/com/example/lab2/Service/Impl/StudentServiceImpl.java
create mode 100644 src/main/java/com/example/lab2/Service/StudentService.java
create mode 100644 src/main/resources/application.properties
create mode 100644 src/test/java/com/example/lab2/Lab2ApplicationTests.java
```

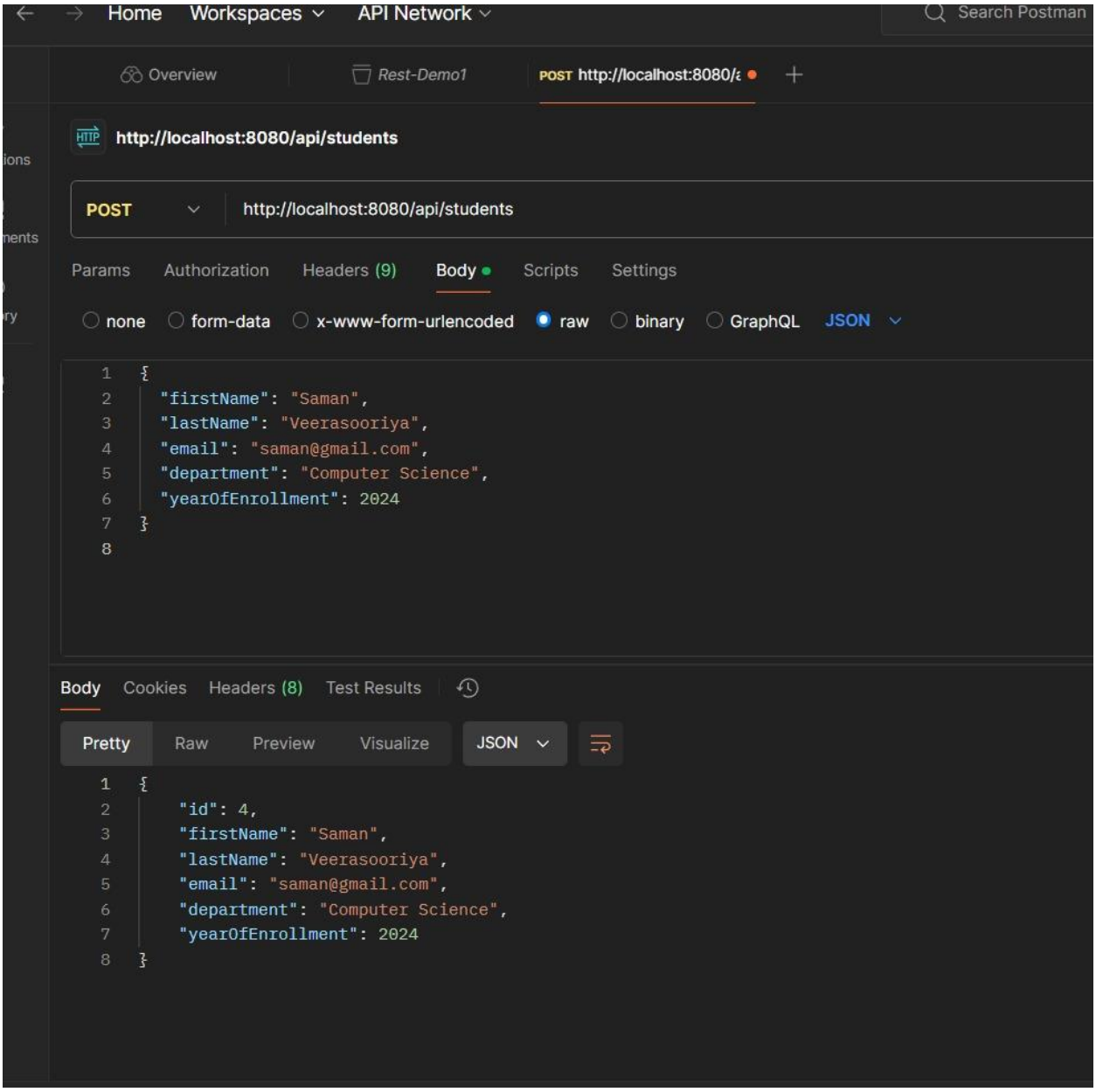
```
D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git branch -M main

D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git remote add origin https://github.com/PasinduPramudithaJ/COSC-32133-labs.git

D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git branch -M main

D:\Users\jayasek-ps20010\Desktop\PS.2020.010-Lab sheet 2\lab2>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 35, done.
Counting objects: 100% (35/35), done.
Delta compression using up to 4 threads
Enumerating objects: 35, done.
Counting objects: 100% (35/35), done.
Delta compression using up to 4 threads
Counting objects: 100% (35/35), done.
Delta compression using up to 4 threads
Delta compression using up to 4 threads
Compressing objects: 100% (25/25), done.
Writing objects: 100% (35/35), 11.11 KiB | 1.11 MiB/s, done.
Total 35 (delta 1), reused 0 (delta 0), pack-reused 0
Writing objects: 100% (35/35), 11.11 KiB | 1.11 MiB/s, done.
Total 35 (delta 1), reused 0 (delta 0), pack-reused 0
Total 35 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/PasinduPramudithaJ/COSC-32133-labs.git
 * [new branch]      main -> main
To https://github.com/PasinduPramudithaJ/COSC-32133-labs.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
branch 'main' set up to track 'origin/main'.
```

Outputs-:


Home
Workspaces
API Network
Search Postman
Overview
Rest-Demo1
POST http://localhost:8080/
http://localhost:8080/api/students
POST
http://localhost:8080/api/students
Params
Authorization
Headers (9)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
{
"firstName": "Saman",
"lastName": "Veerasooriya",
"email": "saman@gmail.com",
"department": "Computer Science",
"yearOfEnrollment": 2024
}
Body
Cookies
Headers (8)
Test Results
Pretty
Raw
Preview
Visualize
JSON
{
"id": 4,
"firstName": "Saman",
"lastName": "Veerasooriya",
"email": "saman@gmail.com",
"department": "Computer Science",
"yearOfEnrollment": 2024
}

http://localhost:8080/api/students/4

PUT http://localhost:8080/api/students/4

Params Authorization Headers (9) Body Scripts Settings

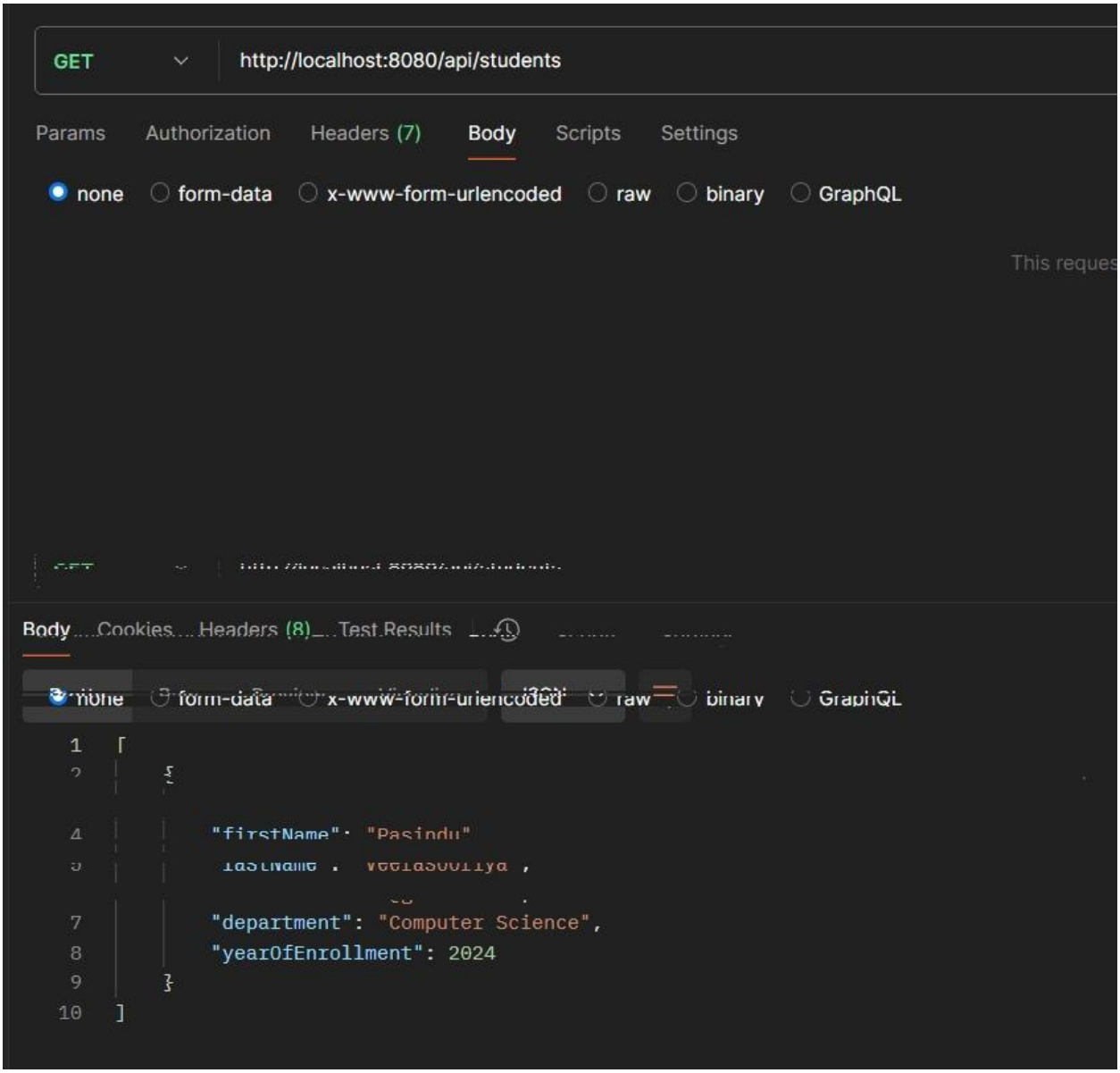
none form-data x-www-form-urlencoded raw binary GraphQL JSON

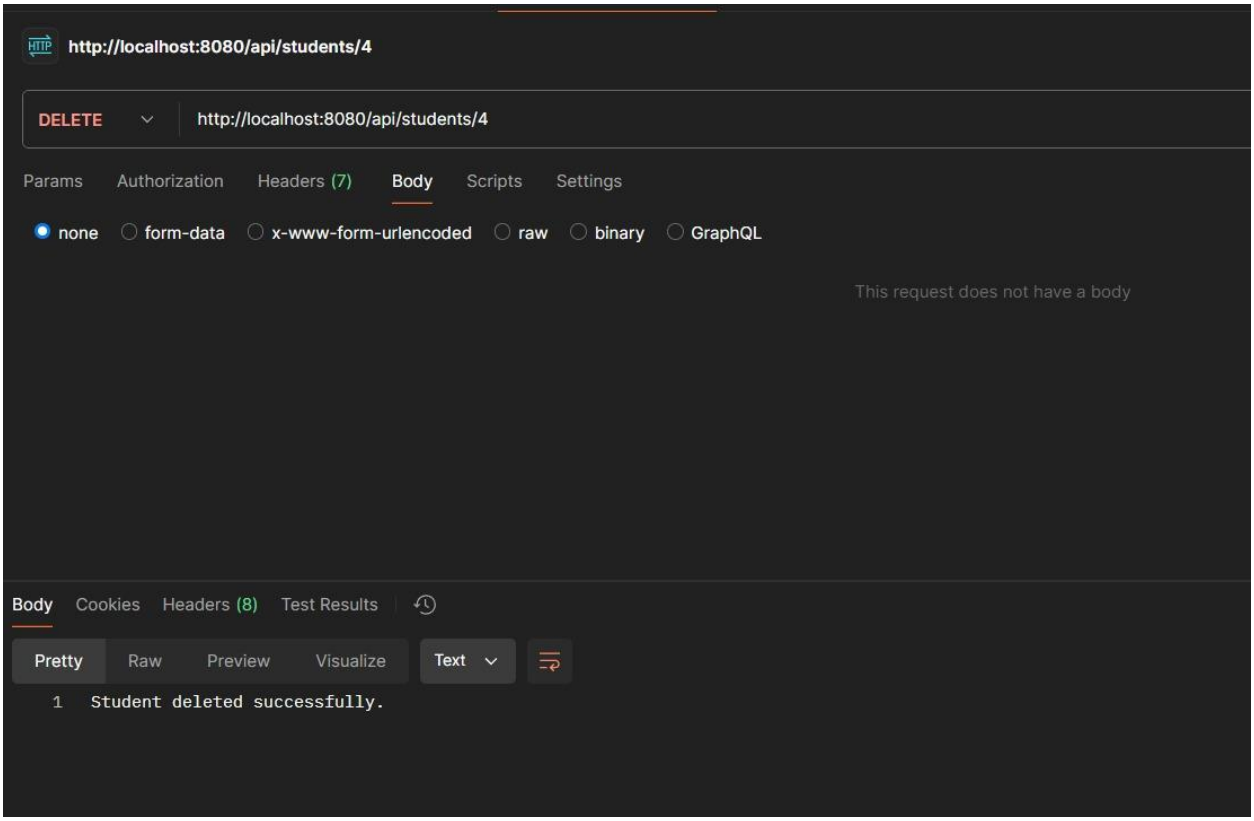
```
{
  "firstName": "Pasindu",
  "lastName": "Veerasooriya",
  "email": "saman@gmail.com",
  "department": "Computer Science",
  "yearOfEnrollment": 2024
}
```

Body Cookies Headers (8) Test Results

Pretty Raw Preview Visualize JSON

```
{
    "id": 4,
    "firstName": "Pasindu",
    "lastName": "Veerasooriya",
    "email": "saman@gmail.com",
    "department": "Computer Science",
    "yearOfEnrollment": 2024
}
```


GET
http://localhost:8080/api/students
Params
Authorization
Headers (7)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Body
Cookies
Headers (8)
Test Results
"firstName": "Pasindu"
"department": "Computer Science",
"yearOfEnrollment": 2024


http://localhost:8080/api/students/4
DELETE
http://localhost:8080/api/students/4
Params
Authorization
Headers (7)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
This request does not have a body
Body
Cookies
Headers (8)
Test Results
Pretty
Raw
Preview
Visualize
Text
1 Student deleted successfully.